EE 121B Fall 2026
Prof. Emaminejad
Homework Assignment 3 Solution

HW3 solution

Problem 1: Step junction

A Si step junction maintained at room temperature under equilibrium conditions has a p-side doping of $N_A = 2\times10^{15}$/cm3 and an n-side doping of $N_D = 10^{15}$/cm3. Compare

a) V_{bi}

b) x_p, x_n, and W

c) ε at x = 0

d) V at x = 0

Make sketches that are roughly to scale of the charge density, electric field, and electrostatic potential as a function of position.

(a)

$V_{bi} = \frac{kT}{q}\ln(\frac{N_A N_D}{n_i}) = 0.614V$

(b)

$W = x_n + x_p = 1.1\times10^{-4}\ cm$

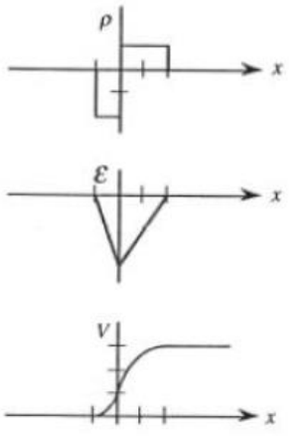
(c)

$\varepsilon(0) = \frac{qN_D}{2\varepsilon_s}x_n = -1.12 \times 10^4\ V/cm$

(d)

$V(0) = \frac{qN_D}{2\varepsilon_s}{x_p}^2 = 0.205\ V$

(e)



Problem 2: pn Junction Diode Current Components

Consider a silicon pn step junction diode with $N_A = 1\times10^{19}$ cm^{-3} and $N_D = 1\times10^{18}$ cm^{-3}, $u_n = 300, u_p = 350$ maintained at $T = 300$K. The minority carrier lifetimes in the p-side and n-side are $\tau_n=10^{-9}$ s and $\tau_p=10^{-8}$ s, respectively.

a) Calculate the minority carrier densities at the edges of the depletion region when the applied voltage (V_A) is 0.6 V.

b) Find the electron and hole current density components as a function of distance on both sides of the junction, for the applied bias voltage of part (a).

c) Calculate the location of the plane (*i.e.* value of *x*) at which the electron and hole current densities are equal in magnitude, for the applied voltage of part (a).

a) Assuming low-level injection, the minority carrier concentrations at the edges of the depletion region are given by: **Lecture 7 slides 11**

$$n_p(-x_p) = \frac{{n_i}^2}{N_A}\left(e^{\frac{qV_A}{kT}}\right) = \frac{10^{20}}{10^{19}}e^{23} = 9.74 * 10^{10} cm^{-3}$$

$$p_n(x_n) = \frac{{n_i}^2}{N_D}\left(e^{\frac{qV_A}{kT}}\right) = \frac{10^{20}}{10^{18}}e^{23} = 9.74 * 10^{11} cm^{-3}$$

b) The diffusion coefficients and minority carrier diffusion lengths are:

$$D_n = \frac{kT}{q}u_n = 7.8\ cm^2/S$$

$$D_p = \frac{kT}{q}u_p = 9.1 cm^2/S$$

$$L_n = \sqrt{D_n\tau_n} = 8.8 * 10^{-5}\text{cm}$$

$$L_p = \sqrt{D_p\tau_p} = 3 * 10^{-4}\text{cm}$$

Hole current in the n-type quasi-neutral region and electron current in the p-type quasi-neutral region are: **Lecture 7 slides 16**

$$J_p(x) = qD_P\frac{{n_i}^2}{N_D L_p}\left(e^{\frac{qV_A}{kT}} - 1\right)e^{-\frac{x-x_n}{L_p}} = 1.6\times10^{-19}\times9.1\times \frac{10^{20}}{10^{18}\times3\times10^{-4}}\boldsymbol{e^{23}}e^{-\frac{x-x_n}{L_p}} = 0.0047e^{-\frac{x-x_n}{L_p}}\ \text{A/cm^2}$$

$$J_n(x) = qD_n\frac{{n_i}^2}{N_A L_n}\left(e^{\frac{qV_A}{kT}} - 1\right)e^{\frac{x+x_p}{L_p}} = 1.6\times10^{-19}\times7.8\times \frac{10^{20}}{10^{19}\times8.8\times10^{-5}}\boldsymbol{e^{23}}e^{\frac{x+x_p}{L_n}} = 0.00138e^{\frac{x+x_p}{L_n}}\ \text{A/cm^2}$$

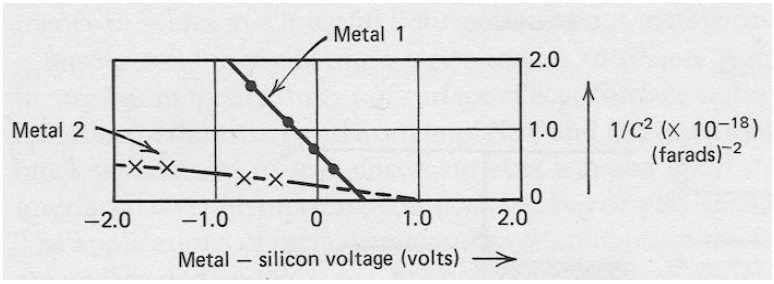
c) In the n-type quasi-neutral region,

$$J_p = J_n = J_{total}/2 = (0.0047+0.00138)/2 = 0.00304 \text{A/cm}^2$$

$$\Rightarrow \quad J_p(x) = 0.0047e^{-(x-xn)/L_p} \Rightarrow (x-x_n)/L_p = 0.436 \Rightarrow (x-x_n) = 1.31 \times 10^{-4} \text{ cm}$$

Problem 3: Small-signal capacitance of a Schottky contact

The following data were obtained for metal contacts to uniformly doped n-type silicon, of equal area:



a) If Schottky theory applies, which metal will probably has the higher work function?

b) Which data correspond to more lightly doped silicon? Justify your answer.

The small-signal capacitance of a Schottky contact to uniformly doped n-type Si is

$$C_J = \frac{dQ_{dep}}{dV_A} = \frac{K_s\varepsilon_0 A}{W} = \frac{K_s\varepsilon_0 A}{\left[\frac{2K_s\varepsilon_0}{qN_D}(V_{bi}-V_A)\right]^{1/2}}$$

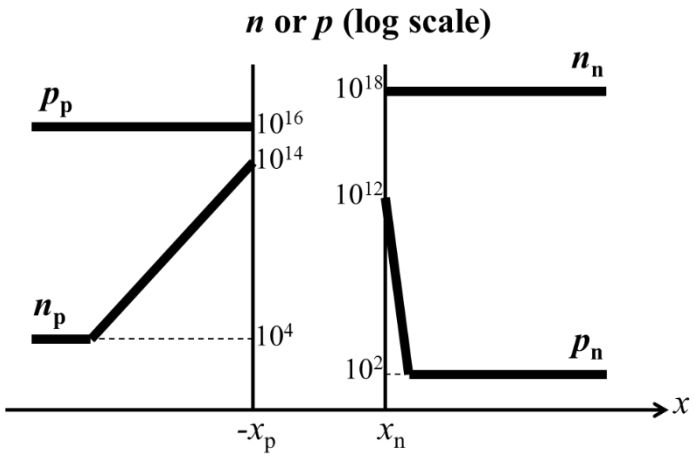
$$\Rightarrow \frac{1}{C^2_J} = \left[\frac{2(V_{bi}-V_A)}{A^2 qN_D K_s\varepsilon_0}\right]$$

a) **Discussion 4 slides 23.** The x-intercept is the built-in potential $V_{bi} = \Phi_{Bn} - (E_c - E_F)$. Since $\Phi_{Bn} = \Phi_M - \chi$, a metal with a larger work function will result in larger V_{bi}. From the plot, V_{bi} for metal 2 is ~1V while V_{bi} for metal 1 is ~0.4V. Since V_{bi} for Metal 2 is larger, **Metal 2 has the larger work function.**

b) **Discussion 4 slides 23.** The slope of the plot is inversely proportional to the dopant concentration, *i.e.* lower N_D corresponds to a steeper slope. Thus, **the contact with Metal 1 corresponds to more lightly doped silicon.**

Problem 4: pn Diode Charge Control Model

The carrier concentrations inside a pn step-junction Si diode with long quasi-neutral regions (much wider than their respective minority carrier diffusion lengths) and cross-sectional area A = 100 μm^2 are plotted below. The minority-carrier lifetimes are $\tau_n = 10^{-6}$ s on the p side and $\tau_p = 10^{-7}$ s on the n side. T = 300 K.



a) Determine the applied voltage V_A.
b) What are the p-side and n-side dopant concentrations?
c) What are the excess minority carrier densities at the edges of the depletion region, *i.e.* $\Delta n_p(-x_p)$ and $\Delta p_n(x_n)$? Do low-level injection conditions prevail in the quasi-neutral regions of the diode? Explain.
d) Calculate the minority carrier diffusion lengths L_n and L_p. (Hint: think of the relationship between doping and mobility)
e) Calculate the excess minority carrier charge stored (Q_P and Q_N) within the quasi-neutral regions.
f) Calculate the diode current using the charge control model. Is it dominated by hole injection into the n side or by electron injection into the p side?

a) Since low-level injection conditions prevail, the "Law of the Junction" holds: within the depletion region and at the edges of the depletion region, $np=n_i^2 \cdot exp\{qV_A/kT\}$.
n_p and p_n each are enhanced by a factor 10^{10} at the edges of the depletion region,
so $10^{10} = exp\{qV_A/kT\} \rightarrow V_A = (kT/q) \cdot ln(10^{10}) = 10 \times (kT/q) \cdot ln(10) = 10 \times (60\text{ mV}) =$ **0.6 V**. **Lecture 7 slides 11.**
b) $p_p = N_A =$ **10^{16} cm^{-3}** and $n_n = N_D =$ **10^{18} cm^{-3}**
c) $\Delta n_p(-x_p) = n_p(-x_p) - n_{p0}(-x_p) = 10^{14} - 10^4 \cong$ **10^{14} cm^{-3}**. $\Delta p_n(x_n) = p_n(x_n) - p_{n0}(x_n) = 10^{12} - 10^2 \cong$ **10^{12} cm^{-3}**
The majority carrier concentrations (p_p and n_n) are not significantly enhanced within the quasi-neutral regions, **so low-level injection conditions prevail**.
d) the electron mobility for $N_A = 10^{16}$ cm^{-3} is $\mu_n = 1200$ cm^2/V·s and the hole mobility for $N_D = 10^{18}$ cm^{-3} is $\mu_p = 150$ cm^2/V·s.
The electron diffusion constant $D_n = \mu_n \cdot (kT/q) = 1200 \times 0.026 = 31.2$ cm^2/s.
The hole diffusion constant, $D_p = \mu_p \cdot (kT/q) = 150 \times 0.026 = 3.9$ cm^2/s.
The electron minority carrier diffusion length

$$L_n = \sqrt{D_n \tau_n} = \sqrt{31.2 \times 10^{-6}} = 5.5 \times 10^{-3} \text{ cm}$$ **= 55 μm**

And the hole minority carrier diffusion length

$$L_p = \sqrt{D_p \tau_p} = \sqrt{3.9 \times 10^{-7}} = 6.24 \times 10^{-4} \text{ cm}$$ **= 0.624 μm**

e) Excess minority carrier charge is stored within the quasi-neutral regions:
Lecture 9 slides 4.

$Q_P = qA\Delta p_n(x_n)\, L_p = 1.6\times10^{-19}\times(100\times10^{-8})\times10^{12}\times 6.24\times10^{-4} =$ **9.98×10^{-17} C** (624 holes

$Q_N = -qA\Delta n_p(-x_p)\, L_n = -1.6\times10^{-19}\times(100\times10^{-8})\times10^{14}\times 5.5\times10^{-3} =$ **8.8×10^{-14} C** (550,000 electrons)

f) The diode current is found using the charge control model:

Lecture 9 slides 6 and 7.

$I_p(x_n) = Q_P/\tau_p = 9.98\times10^{-17}/10^{-7} = 9.98\times10^{-10}$ A

$I_n(-x_p) = -Q_N/\tau_n = 8.8\times10^{-14}/10^{-6} = 8.8\times10^{-8}$ A

$I = I_p(x_n) + I_n(-x_p) =$ **8.9×10^{-8} A**

The current is dominated by electron injection from the more heavily doped n side into the p side.

Problem 5: pn Junction Small-Signal Model

a) Derive the small-signal model for the diode in Problem 4. You may assume that the series resistance of the quasi-neutral regions (R_S) is negligible.

b) Plot $1/C^2$ *vs.* V_A for -2 V < V_A < 0 V. What is the value of the extrapolated *x*-intercept? You may assume that the Boltzmann approximation is valid.

a) **Lecture 9 slides 8, 10 and 11.**

From Problem 4, $I_{DC} = 8.9\times10^{-8}$ A. The small-signal resistance

$$R = (kT/q)/I_{DC} = 0.026/8.9\times10^{-8} = \mathbf{2.9\times10^{5}\ \Omega}.$$

The built-in potential is

$$V_{bi} = \frac{E_g}{2q} + \frac{kT}{q} ln \frac{N_A}{n_i} = 1.12/2 + 0.026\times\ln(10^{16}/10^{10}) = 0.919\text{V}$$

The depletion width

$$W = \sqrt{\frac{2\varepsilon_S(V_{bi}-V_A)}{qN_A}} = 0.2\text{um}$$

Depletion capacitance

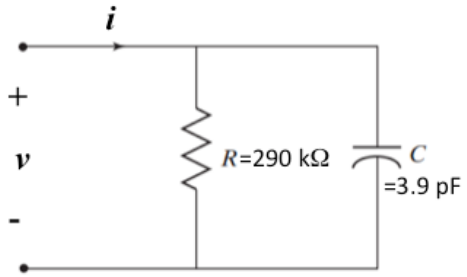
$$C_j = \frac{A\varepsilon_S}{W} = 5*10^{-14}F$$

Diffusion capacitance

$$C_D = \frac{\tau_n}{R} = \frac{10^{-6}}{2.95\times10^{5}} = 0.39\times10^{-11}\ F$$

Total capacitance $C = C_j + C_D =$ **0.39×10^{-11} F**.

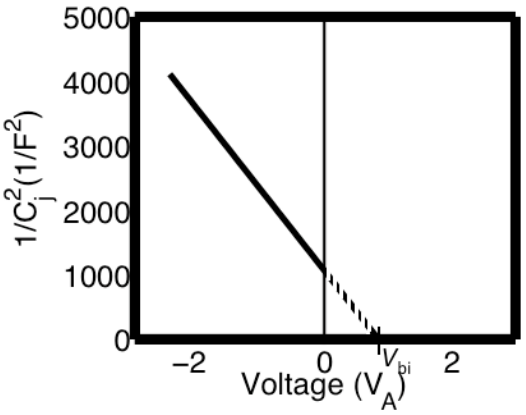
A schematic of the small-signal model is shown below.



b) Under reverse bias, the stored minority carrier charge within the quasi-neutral regions is negligible and so the depletion capacitance (C_J) is the dominant component of small-signal capacitance. From Lecture 13 Slide 14,

$$\frac{1}{C_j^2} = \frac{2(V_{bi} - V_A)}{\varepsilon_s A^2 q N_A} = \frac{2(V_{bi} - V_A)}{10^{-12} \times 10^{12} \times 1.6 \times 10^{-19} \times 10^{16}} = 1.25 \times 10^3 \times (V_{bi} - V_A) \frac{1}{F^2}$$

The plot of $1/C^2$ *vs*. V_A is shown below (Please note that A^2 should be 10^{-12} cm^2 instead of 10^{12} cm^2, so the y-axis should time $1250*10^{24}$ $(1/F^2)$).



Extrapolating to zero, the *x*-intercept occurs at $V_A = V_{bi} =$ **0.919 V**.